

SMART SYSTEM FOR EPILEPTIC SIEZURE PREDICTION AND CONTROL

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EPILEPCY

- "Epilepsy is a serious neurological disorder affecting more than 65 million people worldwide – that is about 1% of the global population.
- Alarmingly, 30% of patients suffer from drug-resistant epilepsy, meaning that medications alone are not effective for them.
- Seizures not only impact quality of life but can also cause injuries, accidents, or even sudden unexpected death in epilepsy (SUDEP).
- Moreover, many seizures are triggered or worsened by stress and fear of having another seizure – creating a vicious cycle for patients."
- (Ref: WHO, Epilepsy Foundation)

WHAT IS epilepsy?

A NEUROLOGICAL CONDITION characterized by *recurrent seizures*

Seizures are due to *brief disturbances* in the *electrical functions* of the brain



Epilepsy affects people of all ages



The cause of
epilepsy for

50%

of people
remains unknown



800,000

Australians will develop
epilepsy during their
lifetime

Around the world
an estimated

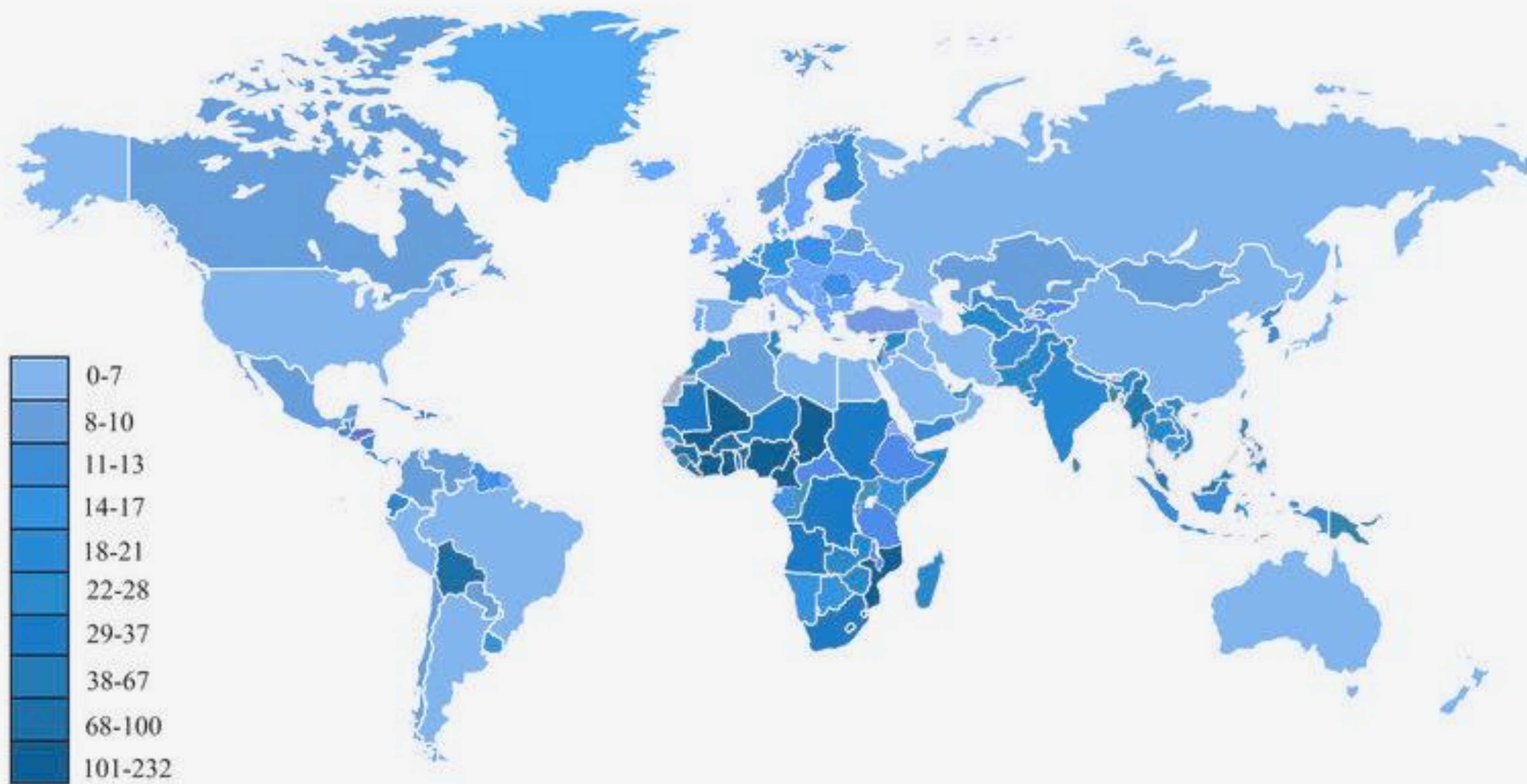
65 million

people have epilepsy
at any given time

1 in 3

people with epilepsy
will not gain full
seizure control with
medication





CURRENT TREATMENTS

- "Today, the most common solution is anti-epileptic drugs (AEDs) such as Levetiracetam, Carbamazepine, or Lamotrigine.
- However, these medications often have side effects and are not effective for one third of patients.
- Implantable Vagus Nerve Stimulation (VNS) devices are FDA-approved and can reduce seizure frequency, but they require invasive surgery and high cost.

IMPACT

- **Health Risks:**

Injuries, accidents, Sudden Unexpected Death in Epilepsy (SUDEP) → up to 17% of epilepsy-related deaths (NIH).

- **Social Impact:**

Patients face psychological stress, fear of recurrence, and social stigma.

- **Economic Burden:**

High cost of long-term medication, hospitalizations, and surgical devices.

Device	Company	Technology	Detection	Prediction	Accuracy	Key Feature
Embrace2 /EpiMonitor	Empatica	EDA, ECG, Motion, Temp	✅ (Tonic-Clonic)	❌	85-90%	FDA Cleared, Low False Alarms
Smartwatches	Apple, Samsung	Heart Rate, Motion	⚠️ Limited (Abnormal Episodes)	❌	70-80%	Readily Available, High False Alarms
NightWatch	LivAssured	Heart Rate, Motion (Armband)	✅ (Nocturnal Tonic-Clonic)	❌	80-85%	Optimized for Night Use, CE Marked
Seer Sense	Seer Medical	cEEG (Behind-ear), ECG	✅ (Generalized)	❌	85-90%	Clinical Grade EEG, CE Marked
NeuroPace RNS System	NeuroPace	Intracranial EEG	✅ (Focal Seizures)	✅ (Prevents Seizures)	90% reduction	Only FDA-approved device for prediction & prevention
Research Prototypes	Universities / Labs	Multimodal + AI	✅	✅ (AI-based)	75-85%	In Development, Not Approved
BioSticker / BioStamp	BioIntelliSense	ECG, Respiration, Motion	⚠️ Limited (General Physiology)	❌	65-75%	FDA-cleared (General Use), Not Seizure-Specific

MARKET GAP

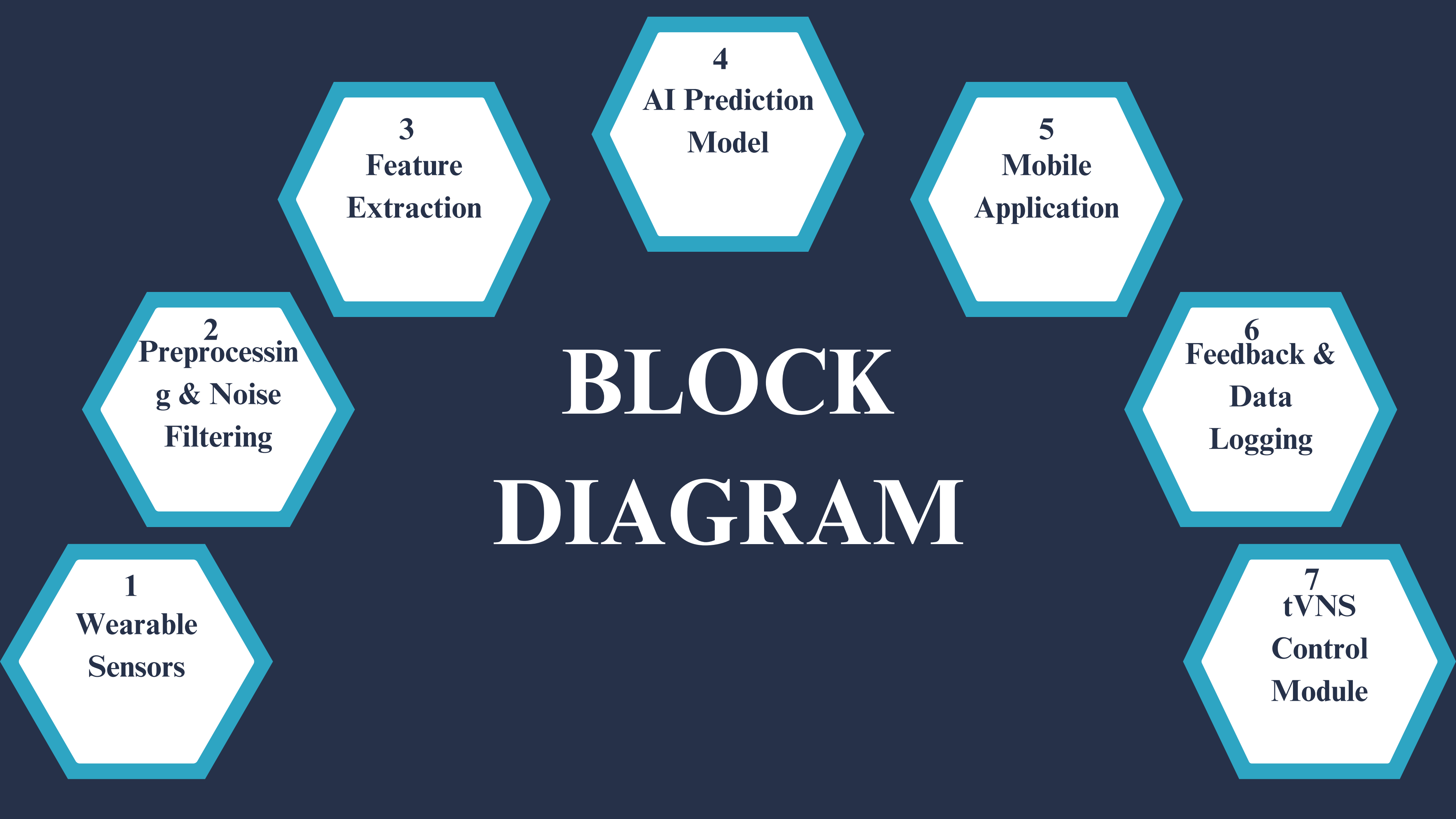
- Existing wearables (e.g., Empatica, EpiWatch, NightWatch) → only detect seizures after onset.
- No affordable, non-invasive system capable of early prediction + active control.
- No integration of AI-based prediction + real-time alerts + tVNS control.

OPPORTUNITY

1. Develop an AI-driven, multimodal wearable solution that integrates:

- Early Seizure Prediction (1_30 min before onset).
- Real-time alerts to caregivers and healthcare providers.
- Non-invasive neuromodulation (tVNS) for potential seizure suppression.

This addresses a critical unmet medical need while opening opportunities in the global Assistive Technology market.



RESOURCES NEEDED & AVAILABILITY

HARDWARE RESOURCES

- **WEARABLE PROTOTYPING KITS:**

Shimmer3 and Empatica E4 → for collecting PPG, EDA, ACC, EMG signals during prototyping and validation.

- **PROCESSING UNITS:**

Raspberry Pi 5 (quad-core ARM CPU, GPU, AI acceleration) → for edge computing and real-time data processing.

ARM Cortex-M4/M7 MCUs → low-power embedded processors suitable for final wearable design.

- **STIMULATION PROTOTYPE:**

Low-cost tVNS prototype built on Arduino + TENS module → to experiment with stimulation parameters (frequency, amplitude, pulse width).

- **CUSTOM PCB DESIGN:**

Integrated board combining sensors + BLE communication + battery management in a compact wearable form factor.

SOFTWARE RESOURCES

- **SIGNAL PROCESSING & MACHINE LEARNING:**

Python (NumPy, SciPy, Pandas, Scikit-learn, TensorFlow/PyTorch) → preprocessing, feature extraction, AI prediction modeling.

MATLAB / GNU Octave → advanced biomedical signal analysis and HRV/EDA indices validation.

- **MOBILE APPLICATION DEVELOPMENT:**

Android Studio (Java/Kotlin) → user interface + BLE communication with wearable.

Flutter (optional) → cross-platform support (iOS & Android).

- **IOT & CLOUD INTEGRATION:**

MQTT protocol / Firebase → secure data streaming, cloud storage, and push notifications.

Bluetooth Low Energy (BLE) → smartwatch-to-app communication.

LAB FACILITIES & TOOLS

- **BIOMEDICAL SIGNAL PROCESSING LAB:**

Equipped with medical-grade ECG/EDA/EMG systems → used for ground-truth data collection and validation.

- **ELECTRONICS WORKSHOP:**

Facilities for PCB fabrication, soldering, oscilloscopes, signal generators, prototyping tools.

DATA SET

1-SEIZEIT2 DATASET

:(OPENNEURO DS005873 V1.1.0)

- Purpose:** First open dataset from a wearable device for focal epilepsy patients.
- Scope:** >11,000 hours of multimodal recordings (EEG, ECG, EMG, motion sensors).
- Participants:** 125 patients (41% female).
- Seizures Recorded:** 886 focal seizures (Avg. duration: 58s, range: 3s – 16min).
- Types:** Focal aware (317), Impaired awareness (393), Focal-to-bilateral tonic-clonic (55).
- Seizure Onset:** Left 44%, Right 12%, Bilateral 1%, Unclear 43%.
- Brain Regions:** Mainly temporal lobe (30%).
- Data Format:** Organized in BIDS standard for research use.

2-CHB-MIT SCALP EEG DATABASE

(PHYSIONET V1.0.0)

3-SIENA SCALP EEG DATABASE (PHYSIONET V1.0.0)

- **Purpose:** Non-invasive EEG dataset for seizure detection/prediction research.
- **Scope:** Scalp Video-EEG + EKG from adult epilepsy patients.
- **Participants:** 14 adults (9 males 25–71 yrs; 5 females 20–58 yrs).
- **Seizures Recorded:** 47 events over ~128 h of data.
- **Seizure Onset:** Expert-annotated start/end times per ILAE criteria.
- **Brain Regions:** 22–29 scalp EEG channels (10–20 system), plus EKG.
- **Data Format:** EDF files (1–5 per subject, ~2 GB max), with metadata includes sampling rate, channel list, seizure timings.

4-SZDB – POST-ICTAL HEART RATE OSCILLATIONS (PHYSIONET V1.0.0)

- **Purpose:** Capture transient low-frequency heart rate oscillations following seizures in partial epilepsy.
- **Scope:** ECG recordings highlighting neuroautonomic responses in the post-ictal period.
- **Participants:** 5 patients with partial epilepsy.
- **Seizures Recorded:** 11 events.
- **Seizure Onset:** Annotated onset/offset times in a separate text file, based on EEG (EEG data not included).
- **Data Format:** WFDB-compatible files—.hea, .dat, .ari, plus times.seize annotations.

METHODS AND THEIR APPROPRIATENESS

1. DATA ACQUISITION AND SIGNAL SELECTION

The system uses a smartwatch to continuously collect multimodal physiological signals relevant to pre-seizure detection. The selected modalities include: Photoplethysmography (PPG), Electromyography (EMG), echocardiography (ECG), and Electrodermal Activity (EDA).

Appropriateness: The selected signals are non-invasive, easily captured via wearable sensors, and supported by strong clinical evidence of pre-seizure changes, making them feasible for continuous monitoring in daily life.

2. AI PREDICTION MODEL

The machine learning model forms the core of the solution and includes the following steps:

- **Data preprocessing:** Noise filtering ,normalization, and segmentation of ra signal
- **Model training:** Supervised learning on patient-specific datasets
- **Prediction period:** Predicting seizure risk within a time period of 30 to 1 minute before seizure onset
- **Personalization:** Continuously adapting to each patient's data stream, improving accuracy over time

Appropriateness: Since epileptic activity is highly individualized, a patient-specific model reduces false positives and enhances sensitivity compared to generic approaches. The adaptive learning framework further ensures robustness and long-term reliability.

3. REAL-TIME ALERTS AND EMERGENCY HANDLING

when an impending seizure is detected, the system activates:

- Patient notifications via smartwatch vibrations and app alerts
- notifications to the physician via the connected mobile app
- emergency protocols in the event of a prolonged seizure or abnormal patterns

Appropriateness: Real-time alerts improve patient safety, reduce the risk of unattended seizures, and reduce response times—critical during sleep or solitary activities.

4. INTERACTIVE DASHBOARD

The mobile app offers two levels of visualization:

- **Patient Dashboard:** Simple, easy-to-read indicators of seizure risk, stress, and sleep quality.
- **Physician Dashboard:** Detailed reports, longitudinal analyses, and downloadable clinical records.

Appropriateness: This dual-interface design enables patients to self-aware and provides physicians with objective, continuous data to improve treatment.

To address behavioral seizure triggers, the app includes: Guided meditation and relaxation exercises, Sleep hygiene recommendations, Stress relief techniques.

Appropriateness: Stress and sleep disturbances are known seizure triggers. By integrating behavioral health tools, the system provides a comprehensive solution that addresses neurological and psychological factors.

The system incorporates a non-invasive T-VNS, using electrodes placed in the ear or neck. Stimulation parameters-frequency, amplitude, and pulse width-are dynamically adjusted according to AI predictions

Appropriateness: T-VNS has demonstrated seizure reduction rates of up to 70%, rates comparable to surgical vagus nerve stimulator, but without the surgical risk and at a lower cost. This ensures easier access and greater patient compliance. T-VNS

DATA FLOW

- Physiological signals are acquired from the smartwatch → preprocessed → transmitted to the mobile application → analyzed by the AI model.
- If seizure risk is predicted, the app triggers alerts and sends stimulation commands to the T-VNS module.
- Data is stored securely in the cloud for visualization, remote monitoring, and model retraining.

CONTROL SYSTEM

CLOED_LOOP TRANSCUTANEOUS VAGUS NERVE STIMULATION T-VNS

What is tVNS?

- Non-invasive electrical stimulation of vagus nerve
- Through ear electrodes (auricular branch)
- Regulates autonomic nervous system
- Modulates brain activity safely

Mechanism of Action:

Increases GABA inhibition

Reduces inflammatory cytokines

Modulates norepinephrine release



TECHNICAL SPECIFICATIONS

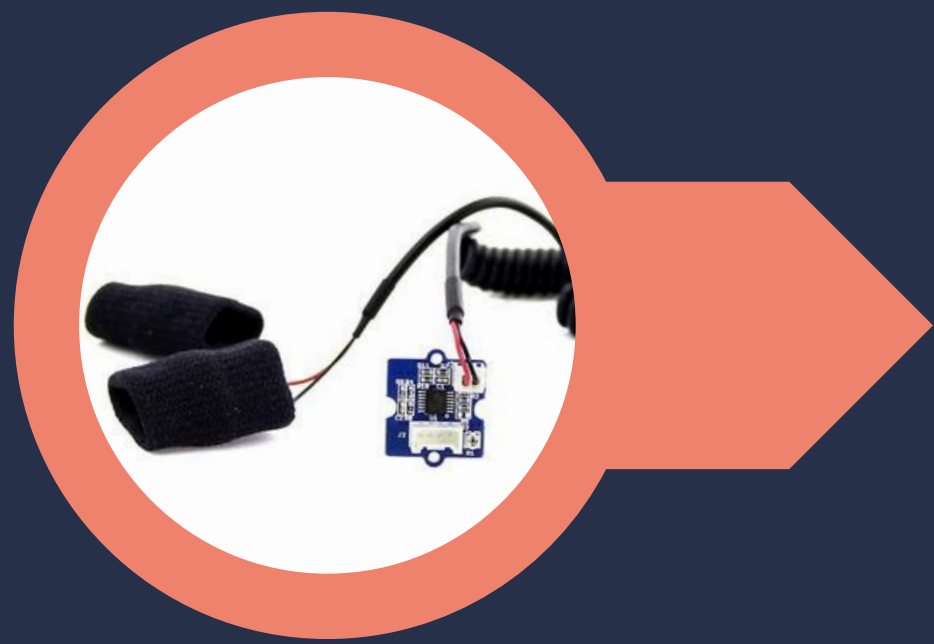
Stimulation Parameters:

- **Frequency:** 20-30 Hz
- **Current:** 1-5 mA (adjustable)
- **Pulse width:** 200-500 μ s
- **Waveform:** Biphasic square wave

Safety Features:

- Current limiting circuit
- Impedance monitoring
- Isolation circuitry
- Emergency stop

HARD WARE



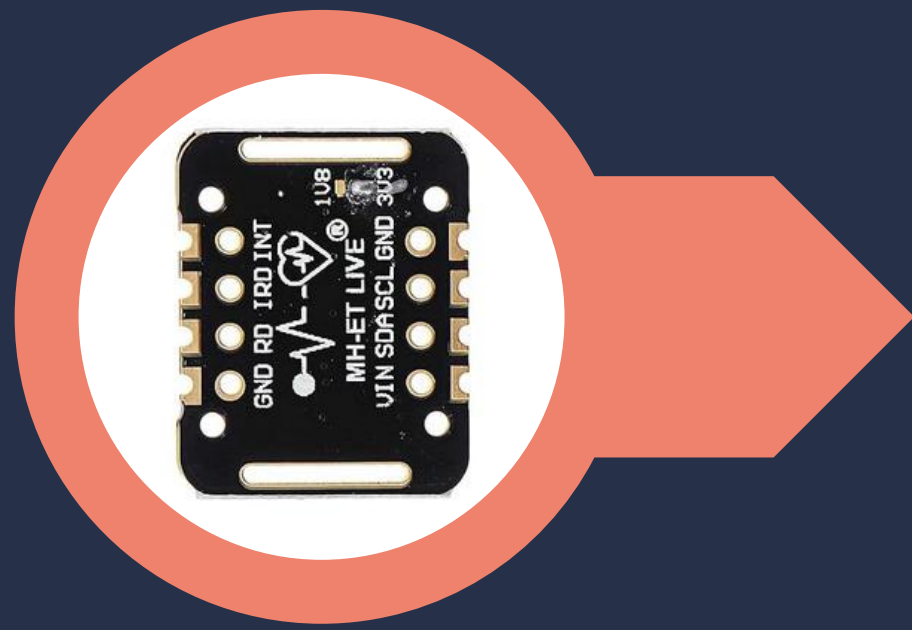
EDA = ELECTRODERMAL ACTIVITY

DEFINITION

HELPS IN SEIZURE DETECTION (SUDDEN RISE IN EDA)

ROLE IN EPILEPSY DEVICES

- CONTROLLED BY THE SYMPATHETIC NERVOUS SYSTEM
- PROVIDES INSIGHTS INTO STRESS AND EMOTIONAL STATES THAT MAY TRIGGER SEIZURES.



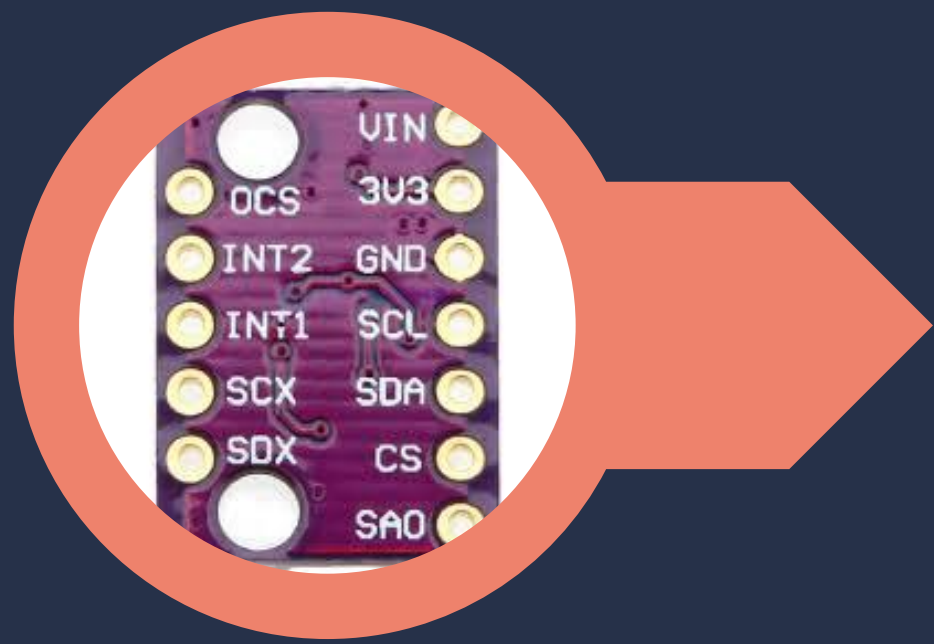
PPG = PHOTOPLETHYSMOGRAPHY

DEFINITION

A NON-INVASIVE OPTICAL TECHNIQUE TO MEASURE BLOOD VOLUME CHANGES IN THE MICROVASCULAR BED OF TISSUE

ROLE IN EPILEPSY DEVICES

- MONITORS HEART RATE (HR) AND HEART RATE VARIABILITY (HRV), WHICH MAY CHANGE DURING SEIZURES.
- PROVIDES ADDITIONAL PHYSIOLOGICAL DATA TO SUPPORT SEIZURE DETECTION.



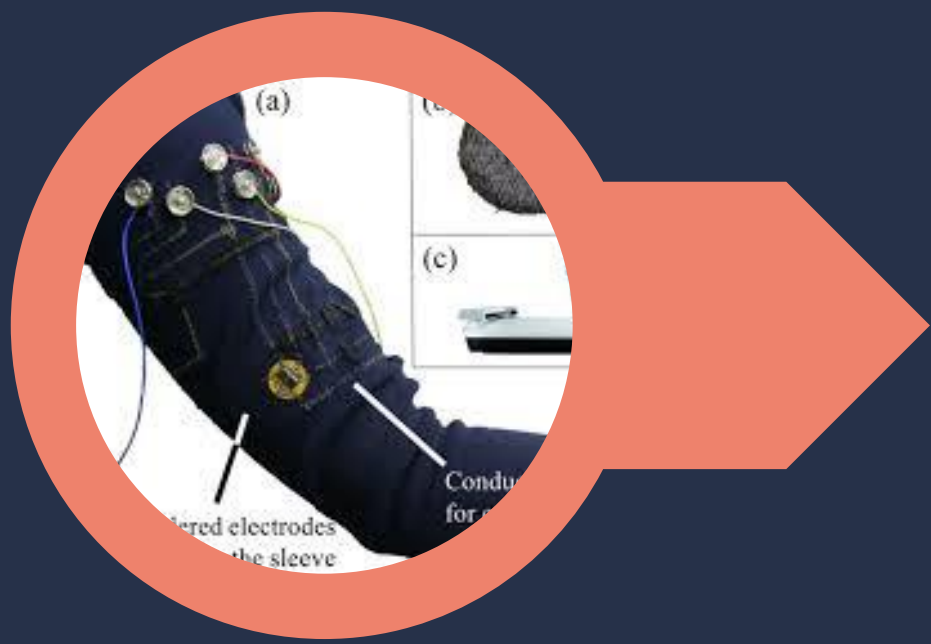
ACC = ACCELEROMETER

DEFINITION

A SENSOR THAT MEASURES MOTION AND ACCELERATION IN 3D

ROLE IN EPILEPSY DEVICES

- IDENTIFIES SEIZURE-RELATED MOTOR ACTIVITY.
- HELPS DISTINGUISH SEIZURES FROM NORMAL DAILY MOVEMENTS



EMG = ELECTROMYOGRAPHY

DEFINITION

EMG MEASURES THE ELECTRICAL ACTIVITY OF MUSCLES

ROLE IN EPILEPSY DEVICES

- HELPS IDENTIFY MOTOR SEIZURES (E.G., TONIC-CLONIC SEIZURES).
- DIFFERENTIATES BETWEEN EPILEPTIC SEIZURES AND NON-EPILEPTIC EVENTS (LIKE FAINTING)

type	sensor	price
E D A	Grove GSR Sensor (Seeed Studio)	(10 – 20)\$
PPG	MAX30102	350 EGP
ACC	LSM6DS3	(1-4)= \$6.43
E M G	Textile EMG	1 7 0 0 E G P
Arduino Nano	Arduino Nano	350 EGP
Electrodes	Ear Silver-clip	3500 EGP
battery	Li-Po pouch 3.7V 250–300 mAh 402530 or 502030	2.50\$
MCU + BLE	nRF52832	3.87\$

Time Line

**Literature review
1 & component
acquisition**

(1 month)

**Software
development &
2 ML training &
Mopile APP**

**Circuit design &
2 PCB fabrication**

(2 months)

**System
3 integration &
testing**

2 months)

**Validation &
4 documentation**

(1 month)

BUSINESS MODEL

TARGET USERS

- Epilepsy patients (especially drug-resistant cases).
- Hospitals & epilepsy centers (for patient monitoring).
- Caregivers & families (for real-time alerts and tracking).

SCALING & GROWTH

- Short Term (Prototype): Arduino/TENS-based safe taVNS for demo & research use.
- Mid Term: Clinical validation with epilepsy patients, increase AI model accuracy, expand dataset.
- Long Term: Regulatory approvals (FDA/CE), scale manufacturing, partner with medtech companies to launch medical-grade taVNS devices.

BUSINESS MODEL

REVENUE STREAMS

DEVICE SALES

- Version 1 (Basic): Smartwatch + Mobile App (Prediction + Monitoring).
- Version 2 (Full): Smartwatch + Mobile App + tVNS Control Device (Prediction + Control).
- Add-on Path: Patients can start with the basic version and later upgrade by purchasing the Control module separately.

SUBSCRIPTION MODEL

Monthly SaaS subscription (\$10/month) for:

- Advanced AI-based prediction updates.
- Cloud analytics & patient history dashboard.
- Caregiver & physician portal.

Partnerships & B2B

- Collaborations with hospitals, insurance companies, and NGOs for bulk adoption.
- Integration into clinical epilepsy care programs.

SWOT ANALYSIS

Strengths (S):

- **AI-based early prediction (30–1 min before seizure).**
- **Non-invasive & wearable (tVNS + smartwatch).**
- **Affordable prototype vs. invasive VNS .**
- **IoT-enabled: mobile app, real-time alerts, location tracking.**

Opportunities (O):

- **Growing Assistive Technology (AT) market.**
- **Potential for clinical trials & hospital partnerships.**
- **Expansion to other neurological disorders (Parkinson's, migraine).**
- **Integration with cloud-based analytics & AI healthcare platforms.**

Weaknesses (W):

- Prediction accuracy depends on multimodal signal quality (EDA, PPG, EMG, ACC).
- Requires continuous data collection & model training.
- Limited clinical validation in early prototype stage.

Threats (T):

- **Regulatory approvals (FDA, CE) required for medical certification.**
- **Competition from established medtech companies (Empatica, NeuroPace).**
- **Possible false alarms affecting user adoption.**

TEAM STRUCTURE

1st

- Responsible for overall system architecture (Wearable + Mobile App + Control).
- Development of AI-based Prediction Model (CNN/LSTM).
- Coordinates integration between hardware, software, and control.

2nd

- Design and integration of Wearable Sensors (PPG, ACC, EMG, EDA).
- PCB design, power management, and prototyping.
- Handles Bluetooth Low Energy (BLE) communication.

3th

- Development of the Mobile Application (Android/Flutter).
- Implementation of signal acquisition & preprocessing pipeline.
- IoT connectivity with Cloud / Edge computing platforms (e.g., Firebase).

4th

- Development of the tVNS Module (non-invasive neuromodulation).
- Configuring stimulation parameters (frequency, amplitude, pulse width).
- Ensures safety validation and compliance with IEC/IEEE medical standards.

REFERENCES

- WHO – Epilepsy facts, 2023.
- Epilepsy Foundation – Drug-resistant epilepsy.
- MDPI, 2023 – HRV/PPG abnormalities before seizures.
- PubMed meta-analysis – EDA pre-seizure.
- Nature, 2022 – EMG/ACC micro-movements.
- Empatica, Apple, NeuroPace official websites.
- gammaCore® clinical trials.